

Subject 1 (12+2 minutes)

Due January 30, 2021 11:19 AM • Closes January 30, 2021 11:20 AM

Instructions

For the given graph, find a (j,g) - walk of minimum weight, which contains each edge of the graph at least once. If the weight of the graph is 500, how much is the weight of the walk you found?

Subject 3 (7+2 minutes)

Due January 30, 2021 11:39 AM • Closes January 30, 2021 11:40 AM

Instructions

(using a calculator is not allowed !!!)

In a text the symbols **a,b,c,d,e,f,g** appear 10, 15, 30, 16, 24, 40 and 35 times, respectively. a) Apply Huffman algorithm for encoding the symbols. b) Calculate the average weighted length of the encoding; c) Decode the sequence 01011110010101110... d) Which problem is solved by Huffman algorithm?

Subject 5 (3+ 2 minutes)

Due January 30, 2021 11:56 AM • Closes January 30, 2021 11:57 AM

Instructions

Find the general term of the sequence $x(n)$ given by the recurrence formula $x(n) = 6x(n-1) - 9x(n-2)$, with $x(1)=6$, $x(2)=5$

Subject 6 (5+ 2 minutes)

Due January 30, 2021 12:07 PM • Closes January 30, 2021 12:08 PM

Instructions

25% of emails in my account are spam. 30% of messages contain the word "offer" but 80% of spam messages contain the word "offer". An incoming email doesn't contain "offer". What is the probability that it is a spam?

Subject 7 (7+2 minutes)

Due January 30, 2021 12:29 PM • Closes January 30, 2021 12:30 PM

Instructions

How many integer solutions has the equation $x_1+x_2+x_3+x_4=21$?

- a) How many of them are positive? (>0)
- b) How many of them satisfy the conditions $0 \leq x_i \leq 7$ for $i=1, 2, 3, 4$?

Subject 8 (15+2 minute)

Due January 30, 2021 12:50 PM • Closes January 31, 2021 12:51 PM

Instructions

(calculator allowed)

At an oral examination the teacher ask questions to a student, one by one, until one of the following event occur: the student gives 3 correct answers or the student gives 4 wrong answers. In the first case, the student is accepted to the written examination. Let p be the probability that a student answers correctly to any of the questions.

- a) Which is the probability that a student is accepted to the written exam?
- b) Let X be the total number of questions asked until a decision (positive or negative). Find $P(X=3)$ and $P(X=4)$ and then write the random variable X and prove that it is correctly defined.
- c) For $p=1/2$, find the expected number of questions asked until a decision.

Subject 9 (4+2 minutes)

Due January 30, 2021 1:04 PM • Closes January 30, 2021 1:05 PM

Instructions

In the previous problem, suppose tha all examined sudents have the same probability $p=1/2$ of answering correctly to any question.

- a) Write the idea expressed by the law of large numbers for the random variable X
- b) If a teacher has to examine 100 students and for each question every student needs 1 minute to give an answer, how much time should he calculate for the whole examination?